

Spatial Pipeline Continuation

1. Geometry Cleaning/Validation

Ran validity checks on all 48 ward polygons from Day 1's `ahmedabad_wards_clean.geojson`:

Check	Result
Invalid geometries	0
Empty geometries	0
Geometry type consistency	All 48 are clean Polygons

Note on the diagonal lines seen in Day 1's plot: investigated further — these are just visual rendering of adjacent wards' shared borders overlapping in the plot, not actual invalid/corrupt geometry. No fix needed.

2. Ward IDs/Names — Already Normalized (Day 1)

Ward_ID (1–48, stable integer) and Ward_Name (cleaned, title-case) were already established on Day 1 and confirmed still valid here.

3. Demographic Join Structure

Built `demographic_join_template.csv` — one row per ward (48 rows), with empty placeholder columns for every demographic feature identified in Khushi's Day 1 research note:

`population_total`, `population_density`, `elderly_pct`, `sc_st_pct`, `literacy_pct`, `outdoor_worker_density`, `slum_density`, `healthcare_access`, `ndvi_green_cover`

Important — no fabricated numbers were entered. Per instructions, if a reliable per-ward figure isn't available, the column stays empty (NA) rather than being filled with a guess.

4. Why the actual numbers aren't filled in yet

The AMC's official ward-wise Census 2011 download (ahmedabadcity.gov.in/SP/Demographics) blocks automated/bot access (confirmed: `robots.txt` disallows it). **This file must be downloaded manually** by visiting the site in a browser and clicking through — it cannot be scripted from this environment.

Action needed: Garima (or whoever's free) visits the site, downloads the actual ward-wise file, and runs it through the `merge_real_demographics()` function in `spatial_pipeline_day2.py`, which handles the join by ward name (with a warning if any names don't match cleanly — AMC's spelling may differ slightly from the DataMeet source's ward names).

5. Files produced

- `demographic_join_template.csv` — the schema, ready to receive real values
- `spatial_pipeline_day2.py` — validation + schema-building code, plus the merge function for when real data arrives

Handoff

- Once real demographic numbers are filled in, this becomes the input to Khushi's Risk Rubric (Day 3–4) and Vrinda's dashboard (Day 4)
- **This is now the #1 blocking task for the spatial/vulnerability side of the project** — someone needs to manually get the real AMC file before Day 3's spatial integration can use real vulnerability numbers instead of empty placeholders